	Case Name: School Sports Facility Development **Project Owner:** Eder Omar Vilca Carrillo	Sector	Construction
OR-AS Operations Research Applications and Solutions	**OR-AS** Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
			One of nine std. scenarios
Processed by	Amin Azimi		User defined distributions
Date	2025	Project Control	Automatic tracking
File Name	C2025-07		Tracking based on user input

1. Project description

Project authenticity

Installation of synthetic turf as part of an educational resource initiative to promote student physical activity and well-being. The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	19
Planned Duration (PD)	45 days
Budget At Completion (BAC)	68,188.69€
Renewable Resources	7 Types
Consumable Resources	0 Types

Network topology	
Serial/Parallel (SP)	66%
Activity Distribution (AD)	83%
Length of Arcs (LA)	25%
Topological Float (TF)	46%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	16.37717	17.46657	2.585106
CRI-rho	16.36798	17.4976	2.84261
CRI-tau	9.526848	12.53619	3.272773

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	6.689224	18.8629	3.344344
CRI-rho	6.689224	18.8629	3.344344
CRI-tau	6.689224	18.8629	3.344344

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	69.68421	42.11895	-0.77508
SI	78.99349	35.93296	-1.41152
SSI	15.56422	17.45225	1.974215
CRI-r	18.78897	16.35038	2.391576
CRI-rho	17.75543	16.92692	2.22185
CRI-tau	11.1664	11.38709	3.058959

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	13.90	-10.54
PV - SPI	16.46	-2.01
PV - SCI	16.74	-1.96
ED - 1	15.16	-11.66
ED - SPI	16.46	-2.01
ED - SCI	16.65	-2.01
ES - 1	8.33	-4.71
ES - SPI(t)	16.23	6.71
ES - SCI(t)	16.50	6.57

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1.14	0.00
CPI	1.48	0.00
SPI	7.66	0.00
SPI(t)	10.61	6.83
SCI	7.91	9.11
SCI(t)	10.83	6.73
0.8 CPI + 0.2 SPI	3.31	8.97
0.8 CPI + 0.2 SPI(t)	3.47	2.42

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over **two** tracking periods. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	21278.40	9247.05	32	1.5	1.15	1.06	0.75
std dev	7206.84	36065.07	226.27	0.16	0.60	0.62	0.35
final	26374.41	34748.91	192	1.38	1.58	1.51	1

2.3.3.2. Time forecasting

PD	45 days

Real Duration	47 days

Delay	4%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	65.50	30.41	45.74	-39.36
PV - SPI	79.00	49.50	74.47	-68.09
PV - SCI	79.00	49.50	74.47	-68.09
ED - 1	64.50	31.82	47.87	-37.23
ED - SPI	79.00	49.50	74.47	-68.09
ED - SCI	79.00	49.50	74.47	-68.09
ES - 1	66.50	27.58	41.49	-41.49
ES - SPI(t)	80.00	46.67	70.21	-70.21
ES - SCI(t)	80.00	46.67	70.21	-70.21

2.3.3.3. Cost forecasting

BAC	68,188.69€

Real Cost	68,188.69€

On Budget	0%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	68188.69	0.00	0.00	0.00
CPI	68188.69	0.00	0.00	0.00
SPI	80943.61	18038.19	18.71	-18.71
SPI(t)	80590.14	17538.30	18.19	-18.19
SCI	80943.61	18038.19	18.71	-18.71
SCI(t)	80590.14	17538.30	18.19	-18.19
0.8 CPI + 0.2 SPI	69903.54	2425.16	2.51	-2.51
0.8 CPI + 0.2 SPI(t)	69871.30	2379.57	2.47	-2.47